Notes on data processing/analysis

Within this folder are a number of scripts I used for preprocessing and statistical analysis of the data we presented in our most recent ICSE manuscript. Before you begin working with this yourselves, there are a handful of things I'd like to mention/clarify:

1. The fMRI Lab at Michigan does some initial preprocessing of the data, including slicetime correction and motion correction. Personally, I don't think slicetime correction is necessary with modern multiband EPI sequences—this was definitely useful in the 'old days' of single-shot EPI and slow TRs, where voxels in neighboring slices could be collected 2-3 seconds apart, but now I'd argue it's irrelevant (that being said, it's certainly not *harmful*). So, I actually re-preprocessed the data myself, skipping slicetime correction and jumping straight into motion correction. You probably won't want to do this, though, since this particular step takes *a lot* of time—it'll probably be easiest to use the slicetime/motion-corrected data provided by the fMRI Lab (NIfTI files prefixed with 'ut') if you're just trying to get the hang of the process as a whole. I opted to do this with our codesynth data just to save myself time and computing hours on our cluster.

2. The fMRI Lab uses a directory structure called BIDS (Brain Imaging Data Structure). This is a fairly recent effort to standardize the organization of fMRI data to enable easy sharing and I'm starting to adapt my own work to it, but for this study, I had reorganized things into a different directory structure that I've used over the years. You'll just need to provide the appropriate path definitions in the code so they fit the format of your data.

3. Statistical analysis of task-related brain activity is multilevel. At the 'first level' (*within-subject*), we define the timing of various *events* or stimulus types within a scan, and then we compute *contrasts* between those conditions. These contrast maps are then carried up into a 'second level' (*between-subjects*) analysis that allows us to make some 'average' inference—this is often as simple as a one-sample *t*-test to identify voxels with activity significantly different from zero. Parameter estimation at the first level requires some way of dealing with the fact that fMRI data have a *nonspherical* noise distribution—that is, model error tends to be correlated both in space and time. Different software packages have their own ways of approaching this problem: FSL nonparametrically *prewhitens* voxelwise timeseries prior to model estimation, while SPM uses a *first-order autoregressive model* to estimate temporally-correlated error in each voxel. Another way to do this, however, is with *robust weighted least squares* (rWLS—the method I've used in both of our ICSE manuscripts). This technique essentially fits a model, identifies scans with unwieldy residual error (as a result of head motion, for example), and then re-fits the model after de-weighting those noisy images (so they have less influence on parameter estimation). If you want to use rWLS, you'll need an additional set of functions for SPM (included here—you just need to stuff it in the 'toolbox' directory within SPM). Otherwise, the default AR(1) approach should include motion parameters as nuisance variables. One more big difference is that AR(1) models should use *spatially smoothed* scans, while rWLS uses *unsmoothed* images (in order to get a better estimate of the 'true' noise at each timepoint—we can then smooth the contrast maps before submitting them to a second level analysis).

4. First-level model specification can also be done in a couple different ways, more generally. You'll see that there are arguments in the code for defining event duration by *reaction time*, or simply using the full stimulus window. We also ran models with *parametric modulation* in order to assess whether activity varies linearly based on the 'difficulty' of each trial. Both of these reference some `.mat` and `.csv` files I created from the Eprime output. If you want to try replicating these analyses, let me know and I can compile all that stuff for you, but it's probably easiest to start with basic task-related contrasts first.

5. Second-level inference requires some additional thresholding for multiple comparisons. The default in SPM12 is to use *familywise error* (FWE) correction, but I don't like this. Basically, it's a cluster-level correction that relies on Gaussian random field theory—RFT makes assumptions about the 'smoothness' of the data that fMRI often fails to meet (see **Eklund et al., 2016** for why this is generally a bad idea). I prefer a voxelwise *false discovery rate* (FDR) correction. This used to be a default option in SPM and I'm not entirely sure why they disabled it, but you can still turn it on by editing line 79 in `spm_defaults.m` (change `defaults.stats.topoFDR` from a `1` to a `0`).

6. Finally—and this might be the scariest part—*don't panic* if you can't perfectly replicate these results. If you opt to use the slicetime/motion-corrected data from the fMRI Lab, it wouldn't surprise me if things look a little different. My lab published a paper a few years ago showing that seemingly-minor variations in preprocessing and the methods we use to deal with motion can have profound effects on the ultimate results (**Turner et al., 2015**). We've also recently noticed nontrivial differences in numerical precision when running the exact same code on the same data with the same software, but on two different operating systems (OS X vs. Ubuntu). And beyond that, there are a host of other experimental factors that may contribute to the replicability of task-related effects (not super relevant for the purposes of repeating an analysis on the same dataset, but you can see **Turner et al., 2018**; **Nee, 2019**; and **Turner et al., 2019** for a discussion of these issues). This is obviously pretty troubling, and it's something the neuroimaging community is still grappling with in order to make fMRI a more robust, reproducible science.

I've almost certainly forgotten some stuff here, but hopefully this is helpful. And please don't hesitate to let me know if you have any questions along the way.